

Almost all loops have multiplication group S_n

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Abstract

Cameron proved that for almost all quasigroups of order n the left translations generate the symmetric or the alternating group, and Häggkvist and Janssen excluded the alternating group; Cameron asked whether the same holds for loops, where the argument breaks because the translation by the identity is trivial. We show that it does, by counting: the probability that the rows of a uniformly random reduced Latin square of order n generate a group not containing A_n is at most $e^{-(\log 2 - o(1))n^2}$, and the group is S_n with probability $1 - O(c^n)$ for some $c < 1$. The inputs are the van der Waerden and Brégman bounds on the number of Latin squares, the Praeger–Saxl bound on the orders of primitive groups, Pyber’s bound on the number of subgroups of S_n , and the Häggkvist–Janssen bound on Latin squares all of whose rows are even permutations. This settles the question, raised by Cameron and recorded by Cavenagh, Greenhill and Wanless as a consequence of their conjecture on the second row, and completes an argument sketched by Eberhard in 2016.

1 Introduction

Let Q be a quasigroup of order n on the set $[n] = \{1, \dots, n\}$, with Cayley table a Latin square L , and for $x \in Q$ let $L_x: y \mapsto xy$ be the left translation, the x -th row of L read as a permutation of $[n]$. The *left multiplication group* $\text{Mlt}_\lambda(Q) = \langle L_x : x \in Q \rangle$ is a transitive subgroup of S_n , and the full multiplication group $\text{Mlt}(Q)$, generated by all left and right translations, contains it. Cameron [1] showed that for almost all quasigroups (that is, for all but a vanishing proportion of the Latin squares of order n) the left translations generate S_n or A_n ; Häggkvist and Janssen [4] then showed that the proportion of Latin squares all of whose rows are even permutations is at most c^n for some $c < 1$, so that A_n never occurs. In particular $\text{Mlt}(Q)$ is 2-transitive, which in Smith’s character theory of quasigroups [10] means that almost all quasigroups have trivial character theory (“rank 2”).

Cameron’s argument is short. A uniformly random Latin square has a uniformly random first row, and by a theorem of Łuczak and Pyber [5] a uniformly random permutation lies in no transitive proper subgroup of S_n other than A_n with probability $1 - o(1)$; since the rows generate a transitive group, that group contains A_n .

For *loops* — quasigroups with an identity element, whose Cayley tables, suitably labelled, are the *reduced* Latin squares, with first row and first column in natural order — the argument is not available: the first row is the identity permutation, and no other row is uniformly distributed. Cameron asked whether almost all loops nevertheless have multiplication group S_n (see [2], and [3, §6], where Cavenagh, Greenhill and Wanless record, as Lemma 6.2, that this follows from their conjecture on the cycle structure of the second row of a random Latin square). The statement is folklore in loop theory, where it is mentioned as an unproved belief. The purpose of this note is to settle it, by an argument that uses no information about the second row at all.

Theorem 1. *Let L be a uniformly random reduced Latin square of order n and $H(L) \leq S_n$ the group generated by its rows. Then*

$$\mathbb{P}[H(L) \not\supseteq A_n] \leq e^{-(\log 2 - o(1))n^2}, \quad \text{and} \quad \mathbb{P}[H(L) = S_n] \geq 1 - 3c^n$$

for some absolute $c < 1$ and all large n . Consequently almost all loops have $\text{Mlt}_\lambda(Q) = \text{Mlt}(Q) = S_n$ and trivial character theory.

The idea, which Eberhard sketched in comments on Cameron's post [2], is that the event is about all $n-1$ non-identity rows simultaneously, and such events are cheap to bound by counting: for a subgroup $K \leq S_n$, at most $|K|^{n-1}$ of the $L(n)/n! = (n!)^{n-1}e^{-n^2+o(n^2)}$ Latin squares with first row the identity have all their rows in K , a proportion of at most $e^{n^2+o(n^2)}/[S_n : K]^{n-1}$, which is $e^{-\delta n^2+o(n^2)}$ as soon as $[S_n : K] \geq e^{(1+\delta)n}$. A transitive proper subgroup lies in a maximal one, and the maximal transitive subgroups of S_n are the wreath products $S_a \wr S_b$, $ab = n$, of which there are fewer than $n \cdot n!$, and the primitive maximal subgroups. The crude count disposes of $S_a \wr S_b$ for $b \geq 3$ (index at least $3^n/n^{O(1)}$) but not of $S_{n/2} \wr S_2$ (index $\binom{n}{n/2}/2 < e^n$) nor of A_n ; for the primitive subgroups one needs a bound on their orders and on their number. We handle every imprimitive group by a single block decomposition (Lemma 2), which is sharper than the crude count — it gives the exponent $\log 2$ in place of $\log 3 - 1$ — and in particular disposes of $S_{n/2} \wr S_2$; the primitive groups by the Praeger–Saxl order bound together with Pyber's bound on the number of subgroups of S_n ; and A_n by the theorem of Häggkvist and Janssen.

2 Preliminaries

Latin squares. Let $\mathcal{L}_n$ be the set of Latin squares of order n on the symbol set $[n]$ and $L(n) = |\mathcal{L}_n|$. We use two classical bounds [11, Ch. 17]: the lower bound $L(n) \geq (n!)^{2n}/n^{n^2}$, a consequence of the van der Waerden permanent inequality applied row by row, and the upper bound $L(n) \leq \prod_{k=1}^n (k!)^{n/k}$, a consequence of Brégman's. In the explicit forms we need,

$$\log L(n) \geq n^2 \log n - 2n^2, \quad \log L(a) \leq a^2 \log a - 2a^2 + a\left(\frac{1}{2} \log^2 a + 2 \log a + 3\right) \quad (1)$$

for all $n, a \geq 1$. The first follows from $n! \geq (n/e)^n$. For the second, $\log k! \leq k \log k - k + \log k + 1$ for $k \geq 1$ gives $(\log k!)/k \leq \log k - 1 + (\log k + 1)/k$, and summing over $k \leq a$ with $\log a! \leq a \log a - a + \log a + 1$, $\sum_{k \leq a} (\log k)/k \leq \frac{1}{2} \log^2 a + 1$ and $\sum_{k \leq a} 1/k \leq \log a + 1$,

$$\sum_{k=1}^a \frac{\log k!}{k} \leq a \log a - 2a + \frac{1}{2} \log^2 a + 2 \log a + 3,$$

which multiplied by a is the claim.

Loops and reduced squares. For $L \in \mathcal{L}_n$ with rows $r_1, \dots, r_n \in S_n$ (so $r_i(c) = L(i, c)$), relabelling the symbols by r_1^{-1} gives the square with rows $\sigma_i = r_1^{-1}r_i$, whose first row is the identity; this map $\mathcal{L}_n \rightarrow \mathcal{L}_n^0 := \{L : r_1 = \text{id}\}$ is $n!$ -to-one, so a uniform L gives a uniform $L^0 \in \mathcal{L}_n^0$, and $|\mathcal{L}_n^0| = L(n)/n!$. Permuting the rows of L^0 so that its first column is $1, 2, \dots, n$ gives a reduced square; each reduced square arises from exactly $(n-1)!$ squares L^0 , so a uniform L^0 gives a uniform reduced square, and the two have the same *set* of rows. A uniform reduced

square is the Cayley table of a uniform random loop on $[n]$ with identity 1, whose left translations are its rows. Hence, writing

$$H(L) := \langle \sigma_2, \dots, \sigma_n \rangle = \langle r_1^{-1} r_i : 2 \leq i \leq n \rangle \leq S_n,$$

the law of $H(L)$ for uniform $L \in \mathcal{L}_n$ is the law of $\text{Mlt}_\lambda(Q)$ for a uniform loop Q of order n , and it suffices to prove Theorem 1 for $H(L)$ with L uniform on $\mathcal{L}_n$ (or, equivalently, L^0 uniform on $\mathcal{L}_n^0$). $H(L)$ is transitive: for $c, s \in [n]$ the symbol $r_1(s)$ occurs in column c in some row i , and then $\sigma_i(c) = s$.

Group theory. We use three facts about subgroups of S_n . A transitive subgroup not containing A_n is either imprimitive, i.e. contained in the stabiliser $\text{Stab}(\Pi) \cong S_a \wr S_b$ of a partition Π of $[n]$ into b blocks of size a with $1 < a < n$, or primitive; a primitive subgroup not containing A_n has order less than 4^n (Praeger and Saxl [7]); and the number of subgroups of S_n is at most 2^{Cn^2} for an absolute constant C (Pyber [8], with $C = \frac{1}{6} \log_2 24 + o(1)$); Roney-Dougal and Tracey [9] have recently shown $2^{n^2/16 + O(n^{3/2})}$.

3 Proof of Theorem 1

Lemma 2 (squares with a block structure). *Let $\Pi = \{C_1, \dots, C_b\}$ be a partition of the columns $[n]$ and $\Pi' = \{S_1, \dots, S_b\}$ a partition of the symbols $[n]$, both into b blocks of size a , $ab = n$. The number of Latin squares $L \in \mathcal{L}_n$ every row of which maps each block of Π onto a block of Π' is at most $(b!)^n L(a)^{b^2}$.*

Proof. Each row r_i of such a square induces a bijection $\pi_i: [b] \rightarrow [b]$ with $r_i(C_j) = S_{\pi_i(j)}$. Fix j, j' and let $I_{jj'} = \{i : \pi_i(j) = j'\}$. In a column $c \in C_j$ each of the a symbols of $S_{j'}$ occurs exactly once, and only in rows of $I_{jj'}$, each of which contributes exactly one symbol of $S_{j'}$ to column c ; hence $|I_{jj'}| = a$, and the subarray with rows $I_{jj'}$ and columns C_j is an $a \times a$ Latin square on the symbol set $S_{j'}$. Thus L is determined by the $n \times b$ array $(\pi_i(j))$, whose rows are permutations of $[b]$, together with one Latin square of order a for each of the b^2 pairs (j, j') , on the row set $I_{jj'}$, column set C_j and symbol set $S_{j'}$ determined by the array (identifying each of these sets with $[a]$ in increasing order). There are at most $(b!)^n$ arrays and $L(a)$ choices for each small square. $\square$

Proof of Theorem 1. Let L^0 be uniform on $\mathcal{L}_n^0$ and $H = H(L^0)$. Since H is transitive, if $H \not\supseteq A_n$ then H is imprimitive or primitive of order less than 4^n .

Imprimitive. Suppose H preserves a partition Π of $[n]$ into b blocks of size a , $1 < a < n$. Then every row of L^0 maps each block of Π onto a block of Π , so by Lemma 2 with $\Pi' = \Pi$,

$$\mathbb{P}[H \leq \text{Stab}(\Pi)] \leq \frac{(b!)^n L(a)^{b^2}}{L(n)/n!}.$$

There are $n!/((a!)^b b!) \leq n!$ partitions Π with blocks of size a and fewer than n divisors a of n . By (1), with $b = n/a$, $(b!)^n \leq b^{nb}$ and $\log n! \leq n \log n$,

$$\begin{aligned} \log \frac{n! \cdot n! (b!)^n L(a)^{b^2}}{L(n)} &\leq 2n \log n + \frac{n^2}{a} \log b + \left(n^2 \log a - 2n^2 + \frac{n^2}{a} \left(\frac{1}{2} \log^2 a + 2 \log a + 3 \right) \right) - n^2 \log n + 2n^2 \\ &= -n^2 \log b + \frac{n^2}{a} \left(\log b + \frac{1}{2} \log^2 a + 2 \log a + 3 \right) + 2n \log n. \end{aligned}$$

If $a \geq \sqrt{n}$, the middle term is at most $n^{3/2}(\frac{1}{2}\log^2 n + 3\log n + 3) = o(n^2)$, so the right side is $-n^2 \log b + o(n^2) \leq -n^2 \log 2 + o(n^2)$. If $2 \leq a < \sqrt{n}$, then $b > \sqrt{n}$, $(n^2/a) \log b \leq \frac{1}{2}n^2 \log b$, and $(\frac{1}{2}\log^2 a + 2\log a + 3)/a \leq \frac{5}{2}$ (the function is decreasing in $a \geq 1$ and is less than 2.32 at $a = 2$), so the right side is at most $-\frac{1}{2}n^2 \log b + \frac{5}{2}n^2 + 2n \log n \leq -\frac{1}{4}n^2 \log n + 3n^2$, which is at most $-n^2 \log 2$ once $n \geq e^{15}$. Summing over Π and a ,

$$\mathbb{P}[H \text{ imprimitive}] \leq n e^{-(\log 2 - o(1))n^2},$$

where the $o(1)$ is $O(\log^2 n/n)$, from $a = n/2$.

Primitive. For a fixed subgroup $K \leq S_n$, the squares $L^0 \in \mathcal{L}_n^0$ all of whose rows lie in K number at most $|K|^{n-1}$. Summing over the at most 2^{Cn^2} subgroups K of order less than 4^n ,

$$\mathbb{P}[H \text{ primitive}, H \not\supseteq A_n] \leq \frac{2^{Cn^2} 4^{n(n-1)}}{L(n)/n!} \leq \exp(-n^2 \log n + (C \log 2 + \log 4 + 2)n^2 + n \log n) = e^{-(1-o(1))n^2 \log n}.$$

Adding the two cases gives the first inequality of the theorem.

The sign. $H \leq A_n$ if and only if every $\sigma_i = r_1^{-1}r_i$ is even, i.e. every row of L has the parity of r_1 . Relabelling the symbols by a transposition is a bijection of $\mathcal{L}_n$ reversing the parity of every row, so this event has probability $2\mathbb{P}[\text{all rows of } L \text{ are even}] \leq 2c^n$ by Häggkvist and Janssen [4]. If $H \supseteq A_n$ and $H \not\leq A_n$ then $H = S_n$; so $\mathbb{P}[H \neq S_n] \leq e^{-(\log 2 - o(1))n^2} + 2c^n \leq 3c^n$ for large n . Finally $\text{Mlt}(Q) \supseteq \text{Mlt}_\lambda(Q) = S_n$ is 2-transitive, which is the rank-2 condition of [10]. $\square$

4 Remarks

Remark 3 (sharpness). The dominant term is $S_{n/2} \wr S_2$, and there the exponent is right: the squares all of whose rows preserve a fixed partition into two blocks of size $n/2$ number $\binom{n}{n/2} L(n/2)^4$ by the bijective version of Lemma 2, and $\binom{n}{n/2} L(n/2)^4 / L(n) = 2^{-n^2 + O(n \log^2 n)}$ by the asymptotics $\log L(m) = m^2 \log m - 2m^2 + O(m \log^2 m)$ that (1) encodes (this counts squares with arbitrary first row; the proportion among those with first row the identity is the same); the union over the $\frac{1}{2}\binom{n}{n/2}$ such partitions changes this by $2^{O(n)}$. The threshold $n \geq e^{15}$ in the proof is an artefact of the case split and has not been optimised: the displayed bound with (1) inserted is already less than 1 for $n \geq 250$ (and equals $e^{-0.88 n^2 \log 2}$ at $n = 1000$), and with the exact van der Waerden and Brégman expressions in place of (1) it is less than 1 for $n \geq 30$ ($e^{-0.37 n^2 \log 2}$ at $n = 30$, $e^{-0.71 n^2 \log 2}$ at $n = 100$).

Remark 4 (quasigroups). With $G = \langle r_1, \dots, r_n \rangle$ in place of H , the same three steps (Lemma 2 with $\Pi' = \Pi$, the count $|K|^n$, and [4]) reprove Cameron's theorem [1] without [5], with the same error bound. The converse direction is not available: the loop case cannot be deduced from the quasigroup case, because the symbol relabelling that produces a reduced square replaces the row group $\langle r_i \rangle$ by $\langle r_1^{-1}r_i \rangle$.

Remark 5 (what is and is not used). Nothing is used about the distribution of any single row. The event $H \leq K$ is the intersection of the $n-1$ events $\sigma_i \in K$, and the identity row, which defeats the one-row argument, contributes nothing to it. The crude count $|K|^{n-1}$ against $(n!)^{n-1} e^{-n^2 + o(n^2)}$ already settles each individual K of index at least $e^{(1+\delta)n}$; the two maximal subgroups of smaller index, A_n and $S_{n/2} \wr S_2$ (with their conjugates), are where the Latin constraints inside K have to be used, through [4] and through Lemma 2 respectively. In particular the theorem says nothing about the conjecture of [3] on the cycle type of the second row, which concerns one row given

the identity row, and which remains open; the consequence recorded in [3, Lemma 6.2] simply does not need it.

Remark 6 (Eberhard’s sketch). In comments on [2] (October 2016), Eberhard observed that the number of Latin squares with all rows in K is at most $|K|^n = (n!)^n/[S_n : K]^n$ against $L(n) = (n!)^n/e^{n^2+o(n^2)}$, concluded that only A_n and $S_{n/2} \wr S_2$ need separate treatment, asked whether the method of [4] adapts to other subgroups, and sketched for $S_{n/2} \wr S_2$ a row-by-row count (fill the first $k \approx n/10$ rows inside the group, bound the rest by the standard row-by-row estimate); Cameron replied that this “should not be too hard” and that the second-row conjecture was the interesting part. The sketch is correct once one passes to maximal subgroups, which takes care of the imprimitive groups with $b \geq 3$; what it leaves open is the primitive case, which needs the order bound of [7] and a count of subgroups such as [8], and the details for $S_{n/2} \wr S_2$. Lemma 2 treats all imprimitive groups uniformly and gives the sharp exponent. As far as we know the argument was not written out.

Remark 7 (small orders). For $n = 5$, of the 56 reduced squares 50 have $H = S_5$ and 6 have $H = C_5$ (the sorted Cayley table of $\mathbb{Z}_5$ for each of the six cyclic subgroups; these are also the all-even ones). For $n = 6$, of 9408 reduced squares 7776 have $H = S_6$, 72 have A_6 and 1560 have an imprimitive H (block sizes 2 and 3, orders 6 to 48). In samples of uniformly random Latin squares of orders 7 to 16 (Jacobson–Matthews chain, 2000–3000 squares per order) every H was S_n or A_n , with A_n in 2.1%, 1.2%, 0.4%, 0.1% of the samples at $n = 7, 8, 9, 10$ and never at $n = 12, 16$.

Remark 8 (labelled and unlabelled loops). The theorem counts labelled loops on $[n]$ with identity 1, i.e. reduced squares. Since almost all Latin squares have trivial autoparatopy group [6], almost all reduced squares have trivial automorphism group as loops, so the proportion of isomorphism classes of loops of order n with multiplication group S_n also tends to 1 (though with a weaker error bound, that of [6]).

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